



Temporal Hidden-State Monitoring for Early Detection of Anomalous LLM Generation

Course Title: Deep Learning

Course Code: 22AIE302

Department : AIE

DYNAMITES



Problem Statement & Objectives

Problem Statement

- Large Language Models can generate incorrect, misleading, or anomalous content. Detecting such behavior only after the complete response may limit the opportunity for early warning.

Motivation

- Monitor internal model behavior during generation
- Identify temporal patterns in hidden-state representations
- Detect anomalous generation as early as possible
- Provide a lightweight risk score
- Keep the underlying LLM frozen

Objectives

- Extract hidden states from GPT-2-medium layers 3–6
- Model token-level temporal evolution
- Combine trajectory-based features with uncertainty measures
- Develop a lightweight detection model
- Evaluate detection at 25%, 50%, 75%, and 100% generation
- Measure detection lead time



Problem Statement & Objectives

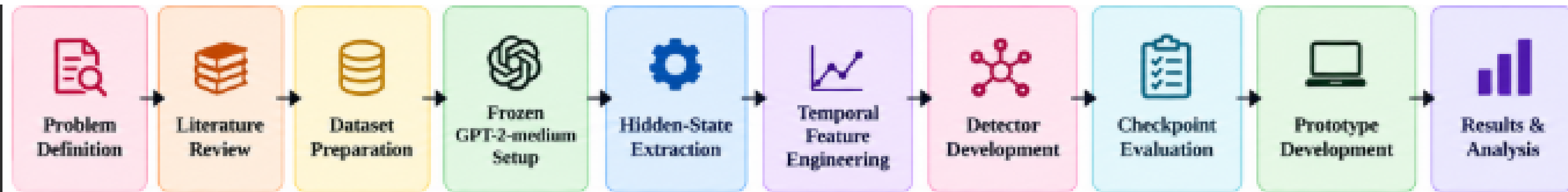
Functional Requirements

- Accept a text prompt as input
- Generate text using frozen GPT-2-medium
- Extract hidden states from Layers 3–6
- Compute temporal trajectory features
- Calculate uncertainty features
- Generate a risk score
- Classify output as NORMAL / SUSPICIOUS
- Provide detection checkpoint and lead-time information

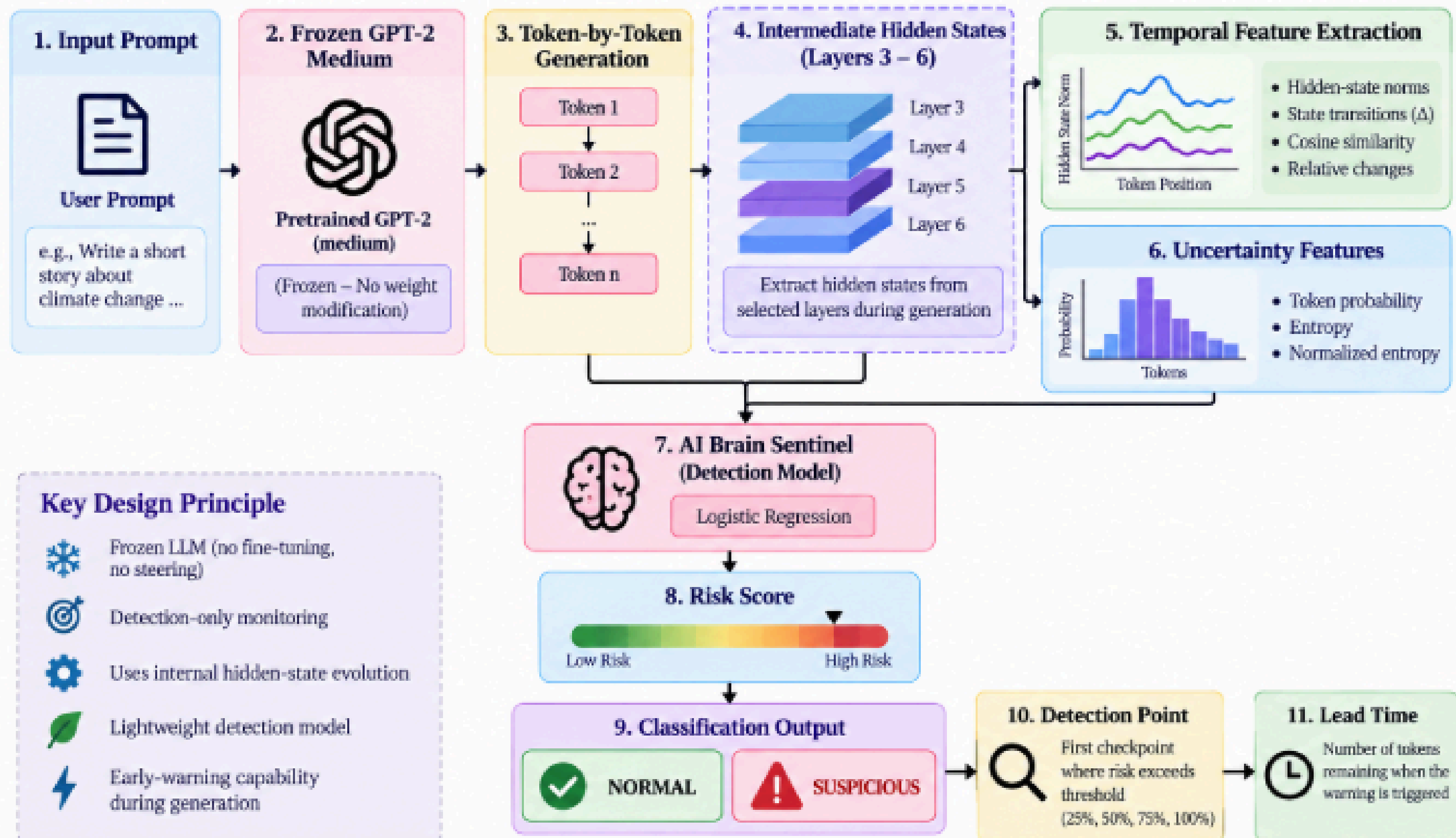
Non-Functional Requirements

- Frozen model with no weight modification
- Lightweight detection model
- Reproducible experimental pipeline
- Efficient token-level monitoring
- Scalable feature extraction
- Detection-only operation

Execution Plan



Proposed System Architecture





Design Specifications & Feature Engineering

Design Specifications

Input

- Text prompt / question
- Frozen GPT-2-medium

Monitored Internal States

- Layers 3, 4, 5, 6
- Hidden-state vector per token

Temporal Features

- Hidden-state norm
- State transition:
- Cosine similarity
- Relative state change
- Cross-layer trajectory statistics

Uncertainty Features

- Token probability
- Entropy
- Normalized entropy

Feature Aggregation

- Mean
- Standard deviation
- Maximum
- Minimum

Detection Checkpoints

- 25% → 50% → 75% → 100%
- The detector evaluates how the internal trajectory evolves as generation progresses.



Prototype / Proof of Concept

Working Prototype

Implemented in Google Colab

Frozen GPT-2-medium used for generation

Real-time token-level hidden-state extraction

Layers 3–6 monitored

Temporal and uncertainty features computed

Logistic Regression detector integrated

Risk score generated at each checkpoint

NORMAL / SUSPICIOUS classification

Detection point and remaining-token lead time calculated

```
First warning: 25%  
Risk score: 0.9228  
Tokens seen: 8  
Tokens remaining: 23
```



Tools & Technologies

Programming & Development

- Python
- Google Colab
- Jupyter Notebook

Deep Learning

- PyTorch
- Hugging Face Transformers
- GPT-2-medium

Machine Learning

- Scikit-learn
- Logistic Regression

Data Processing

- NumPy
- Pandas

Evaluation & Visualization

Matplotlib

ROC-AUC

Confidence Intervals

Cross-validation

Model Configuration

Frozen pretrained model

CUDA / GPU acceleration

No fine-tuning or activation intervention

Initial Results & Findings

	Experiment	ROC_AUC	Std	CI_Lower	CI_Upper
0	Trajectory-only	0.7700	0.0417	0.7335	0.8065
1	Uncertainty-only	0.6030	0.0402	0.5678	0.6382
2	All features	0.7500	0.0499	0.7063	0.7937
3	Early warning - 25%	0.9095	0.0618	0.8553	0.9637
4	Early warning - 50%	0.7625	0.1069	0.6688	0.8562
5	Early warning - 75%	0.7090	0.0933	0.6272	0.7908
6	Early warning - 100%	0.7030	0.0938	0.6208	0.7852



Conclusion

- Developed a detection-only AI Brain Sentinel using frozen GPT-2-medium.
- Monitored temporal hidden-state evolution across Layers 3–6.
- Evaluated trajectory and uncertainty features.
- Achieved the highest controlled-condition ROC-AUC of 0.9095 at the 25% checkpoint.
- Demonstrated an early-warning capability before completion of generation.



Limitations

- Current experiment uses controlled-condition labels, not direct human-verified factual hallucination labels.
- GPT-2-medium is used as the frozen backbone.
- Risk score is not a calibrated probability of factual hallucination.
- Further validation is required across models and datasets.
- Future Work



Future Work

- Evaluate on independently verified response-level labels.
- Test across larger and more diverse datasets.
- Extend to multiple LLM architectures.
- Investigate adaptive detection thresholds.
- Develop a real-time monitoring interface.
- Study generalization across different anomaly types.



THANK YOU